

MAC21 — Last-Minute Prediction Sheet

The questions most likely to appear, by unit, with the exact section they sit in. **Answers are given** where the question has repeated with the same numbers — memorise those outright. Every value below was computed and checked.

Read the repeat counts as evidence, not promises. **4x/6** means it appeared in 4 of the 6 papers carrying that unit (Units I-III: 6 papers · IV: 5 · V: 3). Nothing is guaranteed — but if you only have time for one page, this is the page.

UNIT I — Q1 or Q2

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Expand $xy + \cos(xy)$ about $(1, \pi/2)$, 2nd degree	Q2(c)	7	4x/6	Answer = $\pi/2$ — every derivative vanishes at that point. Show the table of $f, f_x, f_y, f_{xx}, f_{xy}, f_{yy}$ all equal 0, then state $f \approx \pi/2$. Do not panic and hunt for an error. <i>sin variant:</i> $f=1+\pi/2, f_x=\pi/2, f_y=1, f_{xx}=-\pi^2/4, f_{xy}=1-\pi/2, f_{yy}=-1$
Newton-Raphson system $x^2+y^2=x, x^2-y^2=y$, from $(0.8, 0.4)$, 2 iterations	Q1(c)	7	4x/6	$J = \begin{bmatrix} 2x-1 & 2y \\ 2x & -2y-1 \end{bmatrix}$ It1 $\rightarrow x=0.772881, y=0.420339$ It2 $\rightarrow$ $x = 0.771846, y = 0.419644$ $f = 0$ at the start point — that is not an error, $g \neq 0$.
Show $xy(1-x-y)$ is max at $(1/3, 1/3)$	Q2(b)	4	3x/6	r-s-t test. At $(1/3, 1/3)$: $rt-s^2 > 0$ and $r < 0 \rightarrow$ maximum
Open-top box, volume 108 ft^3 , minimum surface area	Q1(d)/Q2(d)	7	2x/6	Minimise $xy+2xz+2yz$ s.t. $xyz=108 \rightarrow$ $x=6, y=6, z=3, S=108$ <i>Open top = ONE xy face, not two.</i>
Negative root of $xe^x - \sin x = 0$, 3 decimals	Q2(b)/(c)	4-7	2x/6	Bracket first: $f(-2.9)=+0.08, f(-3.0)=-0.008 \rightarrow$ start at -3 root = -2.991 $x=0$ is also a root — starting near 0 collapses to it and fails.
Hottest point on unit sphere, $T = 400xyz^2$	Q1(d)	7	2x/6	Lagrange $\rightarrow x^2=y^2=1/4, z^2=1/2 \rightarrow$ $T = 50$

2-markers (pick the easier of the two): geometrical interpretation of Newton-Raphson · define algebraic & transcendental equations · state Maclaurin's series (one variable) · state Taylor's series (two variables) · define maxima and minima of two variables

UNIT II — Q3 or Q4

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Runge-Kutta 4th order	Q3(d)/Q4(d)	7	6x/6	In every paper. $k_1=hf(x,y), k_2=hf(x+h/2, y+k_1/2), k_3=hf(x+h/2, y+k_2/2), k_4=hf(x+h, y+k_3), \Delta y = (k_1+2k_2+2k_3+k_4)/6$ <i>P26 example:</i> $y'=y-x^2, y(2)=5, h=0.1 \rightarrow$ $y(2.1)=5.084146$
Modified Euler's method, 2 corrector iterations	Q4(c)	7	5x/6	Sits at 4(c) in 4 of 5 papers. Predictor $y+hf$, then $y_n + (h/2)[f(x_n, y_n) + f(x_{n+1}, y^*)]$ repeatedly. <i>P26:</i> $y'=-y/(1+x), y(3)=2, h=0.2 \rightarrow$ $y(3.2)=1.9047$
Newton's law of cooling — coffee 92°C , room 24°C , 80°C in 1 min, when 65°C ?	Q3(d)/Q4(d)	7	5x/6	$T = 24 + 68e^{-kt}; k = \ln(17/14) = 0.194156$ $t = 2.606 \text{ min}$ (<i>M24 variant: 90/24/75/60 — same working</i>)
Taylor's series method $dy/dt = e^{-2t} - 2y, y(0)=0.1$, 4th degree	Q3(c)	7	2x verbatim	$y'=0.8, y''=-3.6, y'''=11.2, y^{(4)}=-30.4$ $y(0.1) = 0.163740$
Show $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$ is self-orthogonal	Q4(c)/(d)	7	2x/6	Eliminate λ , then show the DE is unchanged when $y' \rightarrow -1/y'$. That invariance IS the proof — do not solve for λ .
Bungee jumper, $m=68.1, c=0.25$, Euler, 1-second steps	Q4(b)	4	2x/6	$c/m = 0.003671; v: 0 \rightarrow 9.80 \rightarrow 19.25 \rightarrow$ 27.69 m/s
Orthogonal trajectories (Cartesian or polar)	Q3(b)/(d)	4-7	6x/6	Eliminate the parameter FIRST , then swap $y' \rightarrow -1/y'$ (polar: $dr/d\theta \rightarrow -r^2 d\theta/dr$), then solve.

2-markers: DE of closed L-C circuit without e.m.f. $L \ddot{q} + q/C = 0$ · write Euler's formula · define orthogonal trajectories · steps to find orthogonal trajectories · write RK4 formula · two differences between analytical and numerical methods

UNIT III — Q5 or Q6 — your best unit, answer Q5

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
----------	---------	----	------	------------------------

Prove $[1/(D+a)]X = e^{-a \cdot x} \int X e^{a \cdot x} dx$	Q5(b)	4	4x/6	Memorise this outright — it never changes. Let $y = [1/(D+a)]X$, so $(D+a)y = X$, i.e. $y' + ay = X$. Integrating factor e^{ax} gives $(ye^{ax})' = Xe^{ax}$, so $y = e^{-a \cdot x} \int X e^{a \cdot x} dx$
Variation of parameters	Q5(c)/(d)	7	5x/5	In every paper carrying this unit. $W = y_1 y_2' - y_2 y_1'$; $y_1 = -y_1 \int (y_2 R/W) + y_2 \int (y_1 R/W)$ <i>Standard form first — coefficient of y'' must be 1.</i> RHS is always awkward ($\sec^3 x$, $\tan x$, e^{3x/x^2}) — that is the signal.
Cauchy's equation $x^2 y'' + \dots$	Q5(c)/(d)	7	4x/6	Put $x = e^z$: $xy' = Dy$, $x^2 y'' = D(D-1)y$, $x^3 y''' = D(D-1)(D-2)y$. Solve as constant-coefficient in z , then put $z = \log x$. <i>The $-D$ is the most-missed detail in the unit.</i>
Legendre's equation $(ax+b)^2 y'' + \dots$	Q6(c)/(d)	4-7	4x/6	Same as Cauchy with $ax+b = e^z$ and extra factors: $(ax+b)y' = aDy$, $(ax+b)^2 y'' = a^2 D(D-1)y$ <i>Forgetting a^2 wrecks every root.</i>
CF + P.I. for constant coefficients	Q5(c)/Q6(c)/(d)	7	6x/6	Repeated root m (k times) $\rightarrow (C_1 + C_2 x + \dots) e^{mx}$; complex $\alpha \pm i\beta \rightarrow e^{N \cdot x} (C_1 \cos \beta x + C_2 \sin \beta x)$ P.I.: $e^{ax} \rightarrow 1/f(a) \cdot \sin/\cos \rightarrow D^2 \rightarrow -a^2 \cdot e^{ax} V \rightarrow e^{ax} [1/f(D+a)]V \cdot x^n \rightarrow$ expand in <i>ascending</i> powers of D

2-markers: steps for solving Cauchy's LDE **Q5(a)** · define linear and non-linear DE with example **Q6(a)** **3x** · define boundary value problem · write the CF given the roots of the A.E. · define non-homogeneous 2nd-order LDE

UNIT IV — Q7 or Q8

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
Newton's divided difference — cubic through $(2,4) (4,56) (9,711) (10,980)$, find $y(1.5)$	Q7(d)/Q8(c)	7	5x/5	Differences: 1st 26, 131, 269; 2nd 15, 23; 3rd 1 $P(x) = x^3 - 2x$ $y(1.5) = 0.375$ <i>If your expansion has stray x^2 or constant terms, you slipped.</i>
Missing terms — $x = 45, 50, 55, 60, 65$; $y = 3.0, -, 2.0, -, -2.4$	Q8(b)	4	3x/5	3 known values $\rightarrow$ quadratic $\rightarrow$ set both 3rd differences to zero: $3a+b-9=0$ and $-a-3b+3.6=0 \rightarrow$ $y(50)=2.925$, $y(60)=0.225$ <i>Check: 2nd differences all -0.85. One equation is not enough.</i>
Grouped-frequency estimation (income band / age band / marks band)	Q7(c)/Q8(c)	7	3x/5	Build the CUMULATIVE ("less than") column first — that is the whole trick. Interpolate on the cumulative values, then subtract to get the band count. <i>P26: marks 70-75 $\rightarrow y(75)=172.80$, minus 159 $\rightarrow$</i> ≈ 14 students
Simpson / trapezoidal on $\int_0^1 dx/(1+x^2)$	Q7(d)/Q8(b)/(d)	4-7	3x/5	Exact $= \pi/4 = 0.785398$. Ordinates at $h=1/6$: 1, 0.972973, 0.9, 0.8, 0.692308, 0.590164, 0.5 Trapezoidal (4 strips) 0.782794 · Simpson 1/3 0.785398 · Simpson 3/8 0.785396 $\rightarrow \times 4$ gives $\pi = 3.141592$ <i>"Seven ordinates" = 6 intervals. 1/3 needs even n; 3/8 needs n divisible by 3.</i>
Lagrange interpolation (or inverse)	Q7(b)/(c)	4-7	3x/5	$y = \sum y_i \prod_{j \neq i} (x-x_j)/(x_i-x_j)$. Inverse = swap x and y roles. <i>Example: $(-1,9)(0,5)(2,3) \rightarrow$ $P(x)=x^2-3x+5$, $f(1)=3$</i>

2-markers: Lagrange's inverse interpolation formula **Q8(a)** · express $\Delta^2 y_n / \nabla^2 y_n$ in terms of y **Q7(a)** · state Simpson's 1/3 rule · formula for y' , y'' from Newton-Gregory forward · define forward and backward difference

UNIT V — Q9 or Q10 — only 3 papers exist, so 3x = a clean sweep

QUESTION	SECTION	MK	SEEN	ANSWER / WHAT TO WRITE
System of ODEs by matrix method	Q10(d)	7	3x/3	The final question of the exam in all three papers. Eigenvalues of $A \rightarrow$ eigenvectors $\rightarrow X = c_1 V_1 e^{\lambda_1 t} + c_2 V_2 e^{\lambda_2 t}$ $\rightarrow$ apply initial conditions. <i>Rabbit/wolf: $\lambda=2,3$; $r = 360e^{2t} - 120e^{3t}$, $w = 360e^{2t} - 60e^{3t}$</i>
Gauss-Seidel — $x+8y+2z=-4$; $10x+2y+z=59$; $2x-y+20z=74$	Q10(b)/(c)	4-7	3x/3	REORDER FIRST — as printed it is not diagonally dominant. Put the 10x row first, then the 8y row, then the 20z row. It1 (5.9, -1.2375, 3.048125) · It2 (5.842688, -1.992367, 3.016113) Converges to $x=6, y=-2, z=3$

Diagonalization and hence A^n	Q9(c)/(d)	7	3×/3	P = eigenvectors as columns, D = eigenvalues in the same order ; $A^n = PD^nP^{-1}$ $A = \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} : \lambda = 2, 3, P = \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix},$ $A^9 = \begin{bmatrix} -6049 & -6305 \\ 12610 & 12866 \end{bmatrix}$
Rayleigh's power method	Q9(c)/(d)	4–7	3×/3	$Y = AX$, factor out the largest component, repeat. TRAP: on $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ from $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T$ the first iterates go 1, 2, -4 — wild. It needs ~10 iterations to reach $\lambda = 2 + \sqrt{2} = 3.4142$. If it is still jumping, keep going and say so.
Gauss elimination word problem (dietician / electrical network)	Q9(d)/Q10(c)	7	2×/3	Write the system from the table by row (one nutrient = one equation), then row-reduce. Dietician: $x=0.625, y=3.125, z=5 \text{ grams}$
Rank / consistency of $AX=B$	Q9(b)/Q10(c)	4–7	2×/3	$\text{rank}(A) \neq \text{rank}(A B) \rightarrow$ inconsistent · $=n \rightarrow$ unique · $=r < n \rightarrow$ $n-r$ free parameters. For $x+2y-z=a; 2x+y+z=b; x-y+2z=c$ the condition is $b = a+c$

2-markers: define rank · define row echelon form · consistency conditions for $AX=B$ · define modal and spectral matrix · is every square matrix diagonalizable? (**no** — needs n independent eigenvectors) · geometrical interpretation of infinitely many solutions (the two lines coincide)

In the hall

- Read all ten questions (5 min). In each unit pick the side where you can start *more parts*.
- All five 2-markers first** — 10 marks in 15 minutes while you are calm.
- All five 4-markers — you are at 30 marks by minute 50.
- Then the 7-markers, easiest unit first, ~24 min each.
- Never leave a part blank.** Formula + setup + first two steps is routinely 3 of 7 for ninety seconds. Five partial attempts beat two perfect answers and three blanks.
- Show every table. These are method-marked — a Gauss-Seidel or RK4 table with a slip in the last row still scores most of the marks.

Compiled from 6 papers plus the official model paper and course handbook. Repeat counts are out of the papers carrying each unit (I–III: 6 · IV: 5 · V: 3). All numerical answers were computed and verified. Predictions are inferences from observed frequency and slot position, not guarantees — read both options before committing.